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DELIVERABLE 2 – Selected Field

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FIELD DEFINITION

- **Level 1 (Macro Field):** Digital Health & Tele-rehabilitation.
- **Level 2 (Specific Domain):** Wearable Inertial Sensors & Edge Machine Learning for Neurodegenerative Disease Monitoring.
- **Level 3 (Manageable Focus / Selected Field):** Adaptive baseline anomaly detection and closed-loop emergency response systems for early Freezing of Gait (FoG) intervention and fall management in Parkinson's Disease patients.

Field Scope Description

The selected field addresses the intersection of continuous biometric tracking and real-time edge computing for neurological mobility disorders. Rather than using static thresholds or population-wide movement averages, the focus lies on personalized, adaptive machine learning models that learn individual gait baselines (accounting for pharmacological ON/OFF fluctuations). The system provides closed-loop sensory cueing (acoustic/haptic) during pre-freezing states and features an integrated, low-false-positive emergency safety net that dispatches real-time geolocation and pre-impact kinematic black-box data to caregivers and clinicians upon fall detection.

RELEVANCE

Innovation Potential

- **Personalized Baseline vs. Static Rules:** Existing solutions use generic thresholds that cause frequent false alarms. Introducing an adaptive model that learns the specific patient's movement profile significantly increases reliability and real-world adoption.
- **Proactive Prevention vs. Passive Alerting:** Most commercial devices only detect falls after they happen. This approach anticipates gait freezing moments before it occurs, using real-time sensory cues (vibrations/sounds) to help the patient keep walking safely.
- **Objective Clinical Insight:** Recording movement data right before an incident gives neurologists an objective view of motor degradation and medication efficacy, replacing subjective patient diaries.

Social and Environmental Impact

- **SDG 3 (Good Health and Well-Being):** Preserves independence and mobility for Parkinson's patients, reducing fear of falling, isolation, and dangerous injuries (such as hip fractures).
- **SDG 9 (Industry, Innovation, and Infrastructure):** Drives modern digital health adoption through smart, decentralized monitoring that securely handles patient data locally.
- **SDG 10 (Reduced Inequalities):** Enables continuous, high-quality motor monitoring at home, making specialized neurological follow-up accessible even to patients far from major clinical centers.

Trend Analysis and Market Demands

- **Aging Demographics:** Parkinson's is among the fastest-growing neurodegenerative conditions globally, driving an urgent need for solutions that prevent costly emergency hospitalizations.
- **Shift to Remote Care (Hospital-at-Home):** Healthcare systems are actively investing in remote patient monitoring and tele-neurology to decrease hospital stays and optimize outpatient therapy.
- **Rise of Smart Wearables:** Patients and doctors increasingly expect accessible, connected health devices that provide actionable feedback without requiring complex clinical setups.